

TASK 1 — ML INTERNSHIP PROGRAM

Uber Fare Prediction

Exploratory Data Analysis



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Project Objective

Understand the dataset before Machine Learning.



Fare Factors

Explore variables influencing amounts.



Temporal Trends

Analyze ride demand fluctuations.



Data Quality

Cleanse invalid and duplicate records.



Extract Insights

Guide the next modeling stage.

Data Cleaning & Quality

INITIAL DATASET

500,000

Rows × 26 columns

FINAL DATASET

487,742

Cleaned rows ready for modeling

Main Cleaning Steps

- ✓ Handled missing values.
- ✓ Removed duplicate records.
- ✓ Filtered invalid fare amounts.
- ✓ Dropped zero passenger-count records.
- ✓ Cleaned abnormal geographical coordinates.
- ✓ Removed extreme distance values.
- ✓ Excluded suspicious high-fare observations.

QUESTION 01

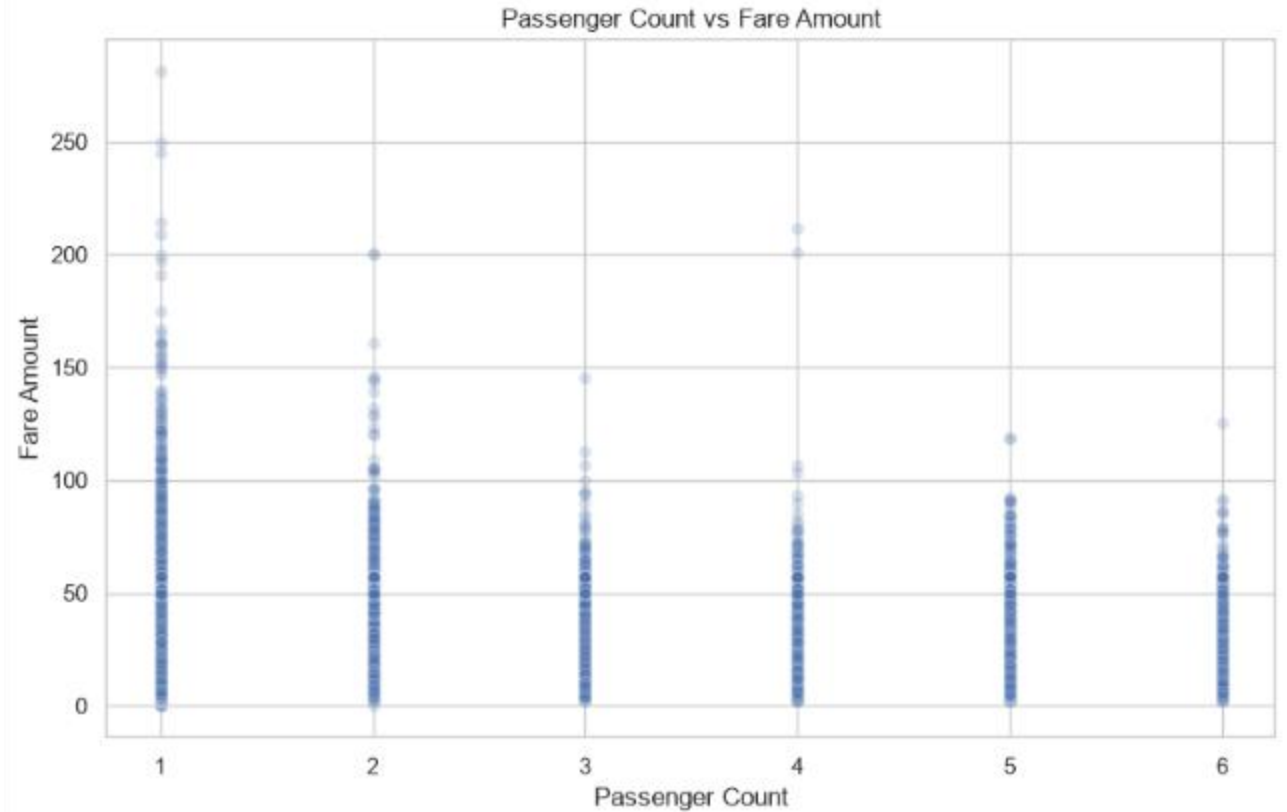
Does a higher passenger count lead to a higher fare amount?

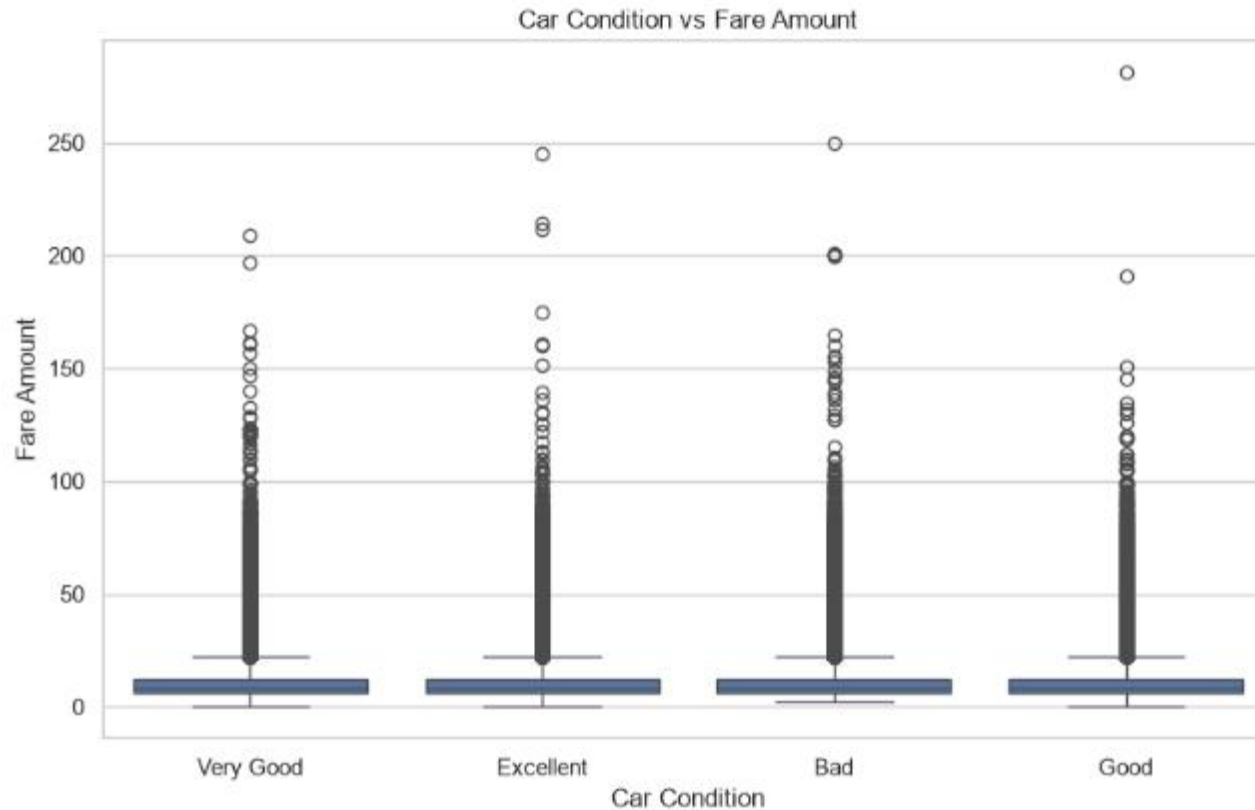
Key Insights

No clear increasing relationship observed.

Average fares stable across passenger counts.

Weak individual predictor of fare.





QUESTION 02

Does car condition influence fare amount?

Key Insights

Distributions highly similar across conditions.
Average fares tightly clustered together.
No apparent strong relationship observed.

QUESTION 03

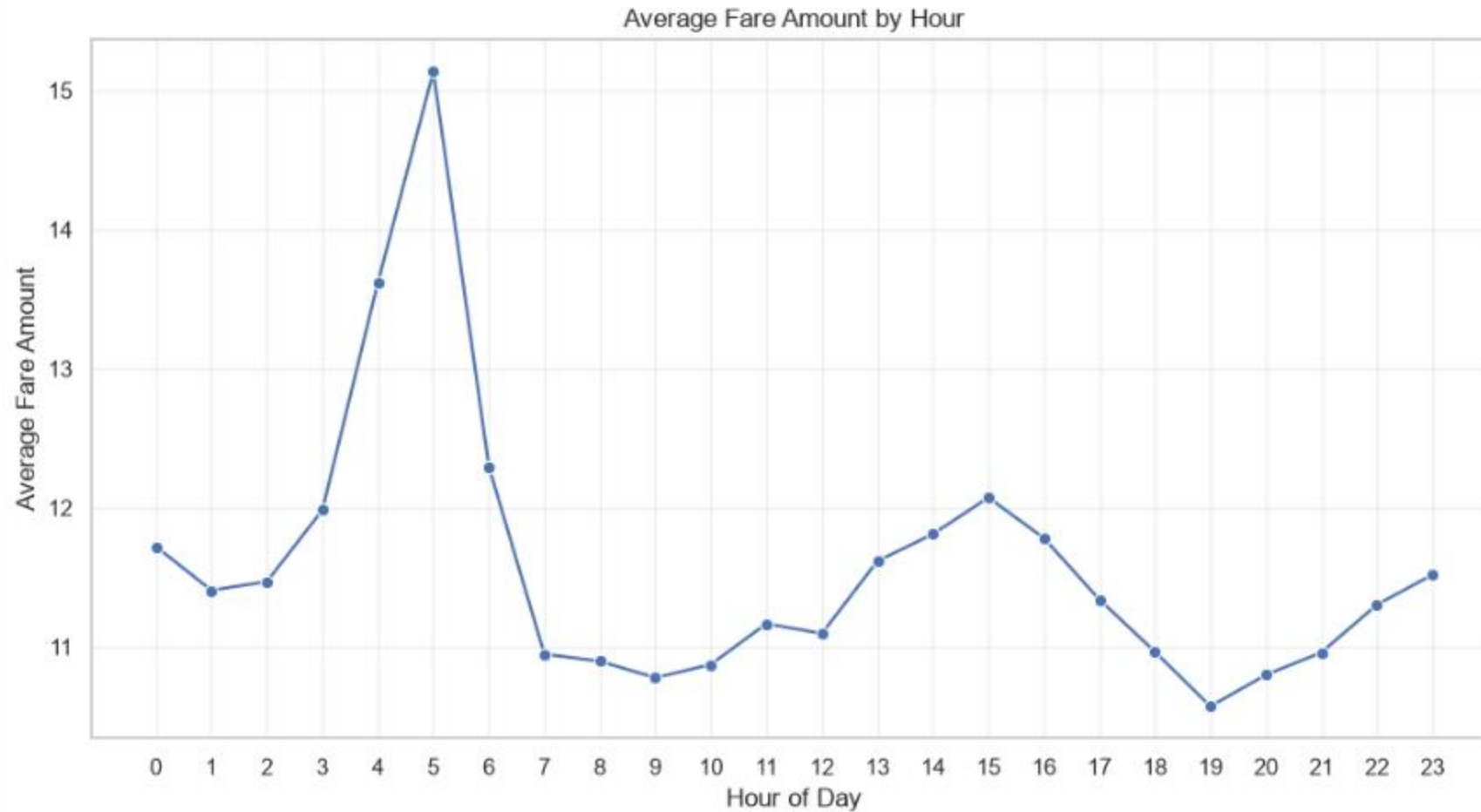
At what hour of the day are fares typically highest?

PEAK FARE TIME

AVERAGE PEAK FARE

5:00 AM

\$15.14



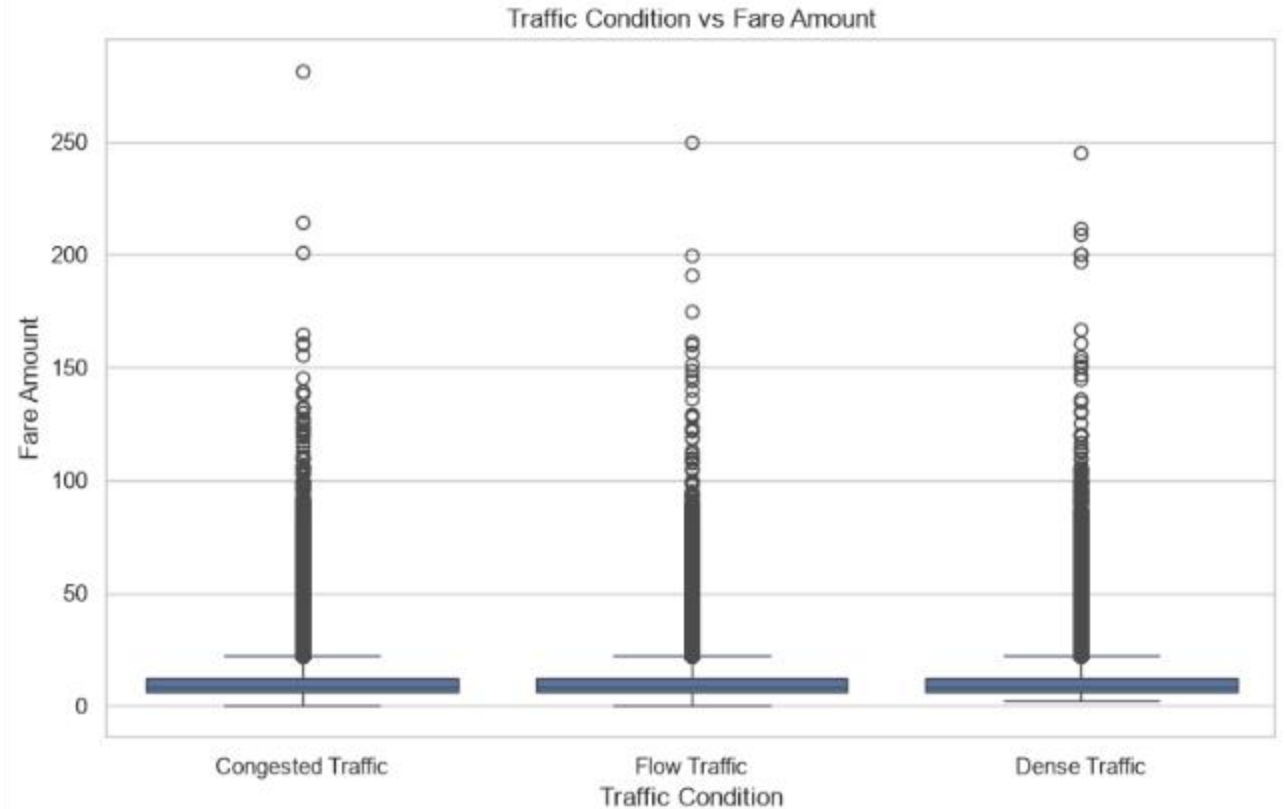
QUESTION 04

Does traffic condition affect trip fare?

Key Insights

Average and median fares identical across states.

Traffic condition shows minimal baseline impact.





QUESTION 05

Is trip distance strongly related to fare?

CORRELATION COEFFICIENT

0.838

Key Insights

Strong positive linear relationship.

Strongest numerical indicator examined.

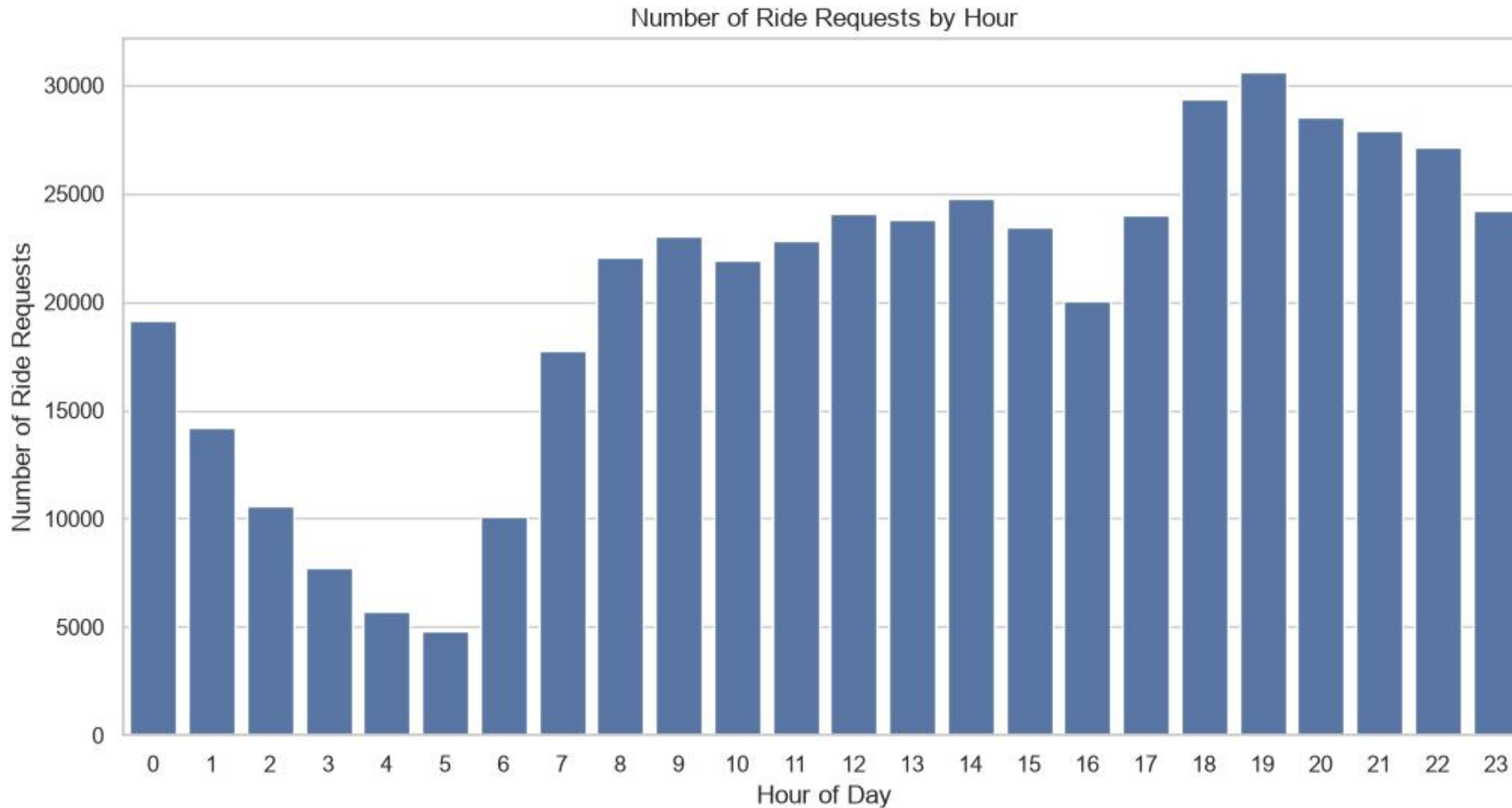
Longer trips directly correspond to higher fares.

QUESTION 06

At what time of day are ride requests most frequent?

PEAK DEMAND TIME TOTAL REQUESTS

7:00 PM 30,602



QUESTION 07

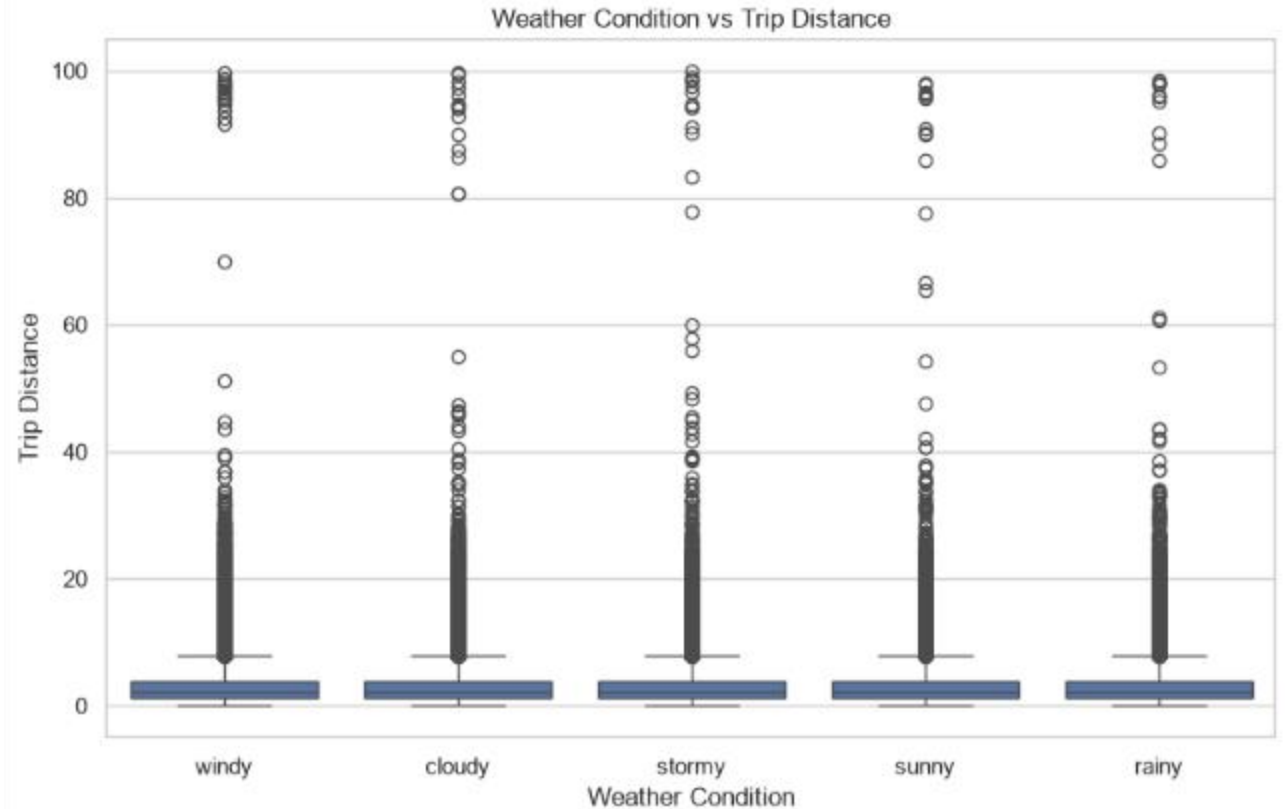
Does weather condition influence average trip distance?

Key Insights

Distances highly similar across all conditions.

Minimal variance between weather categories.

No strong apparent weather influence detected.

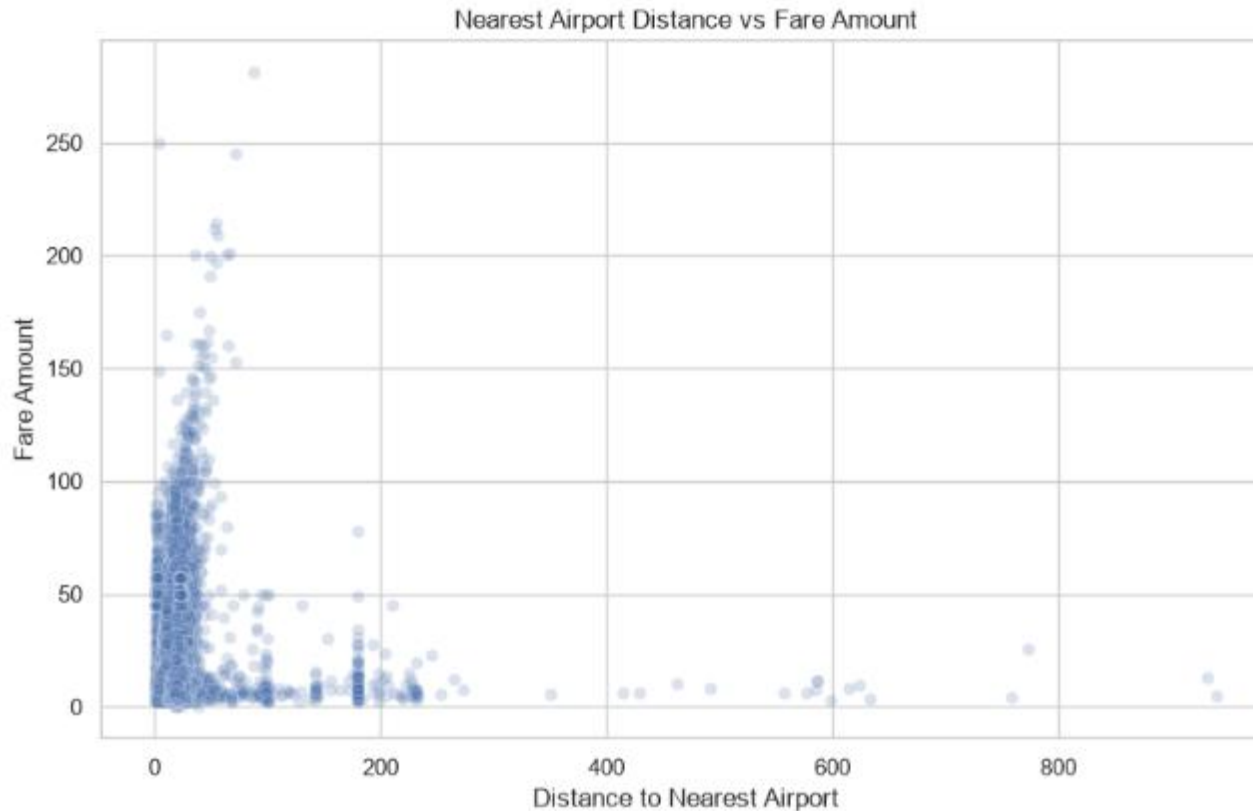


QUESTION 08

Are rides closer to airports generally more expensive?

CORRELATION COEFFICIENT

-0.031



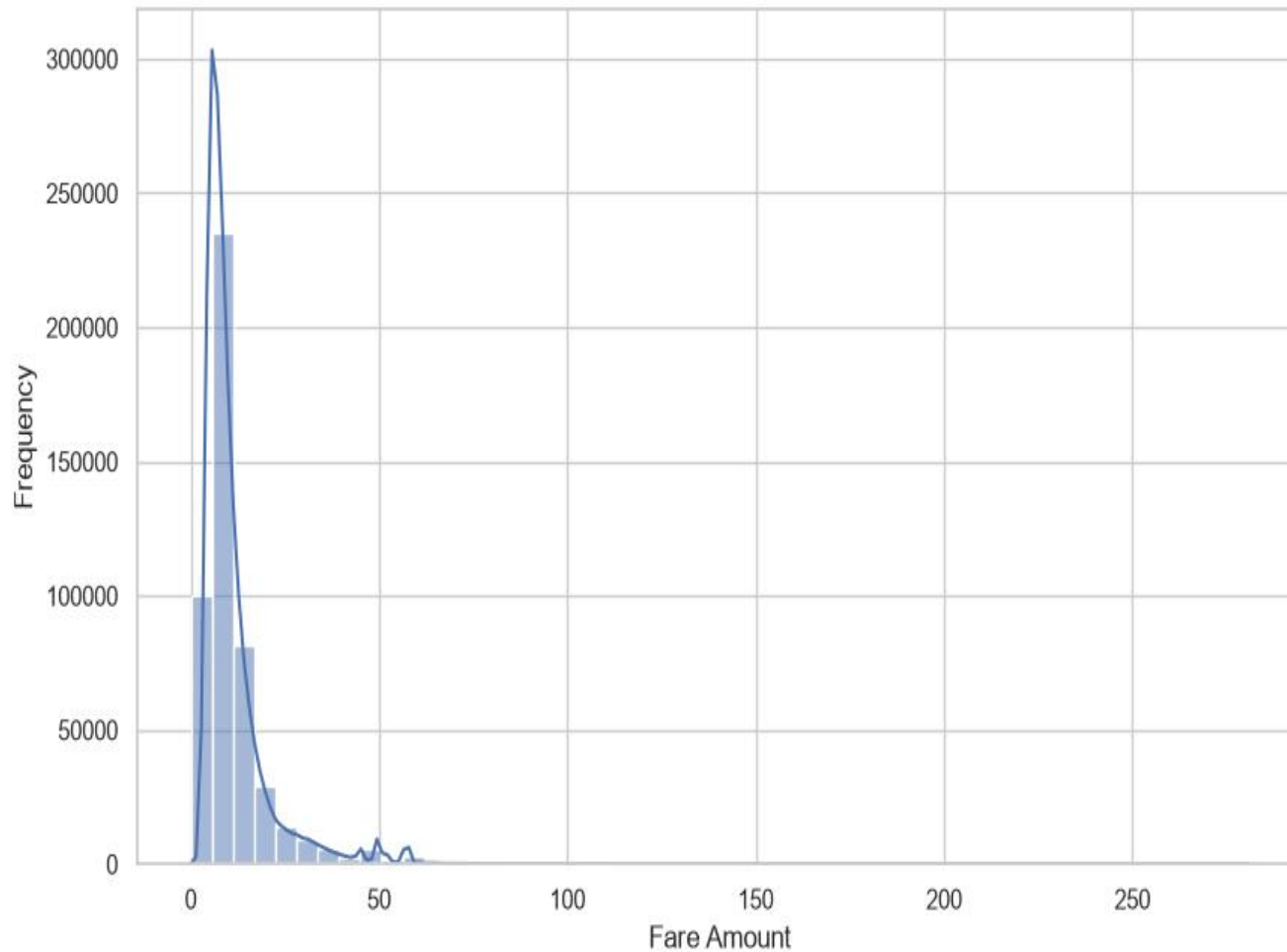
Key Insights

Correlation approaches zero.

No linear relationship observed.

Airport distance fails to explain fare variance.

Distribution of Fare Amount



QUESTION 09

How is the Fare Amount distributed?

MEAN

\$11.32

MEDIAN

\$8.50

90% OF FARES

≤ \$20.50

MAXIMUM

\$281

Key Insights

Fares heavily concentrated below \$20.

Highly right-skewed distribution.

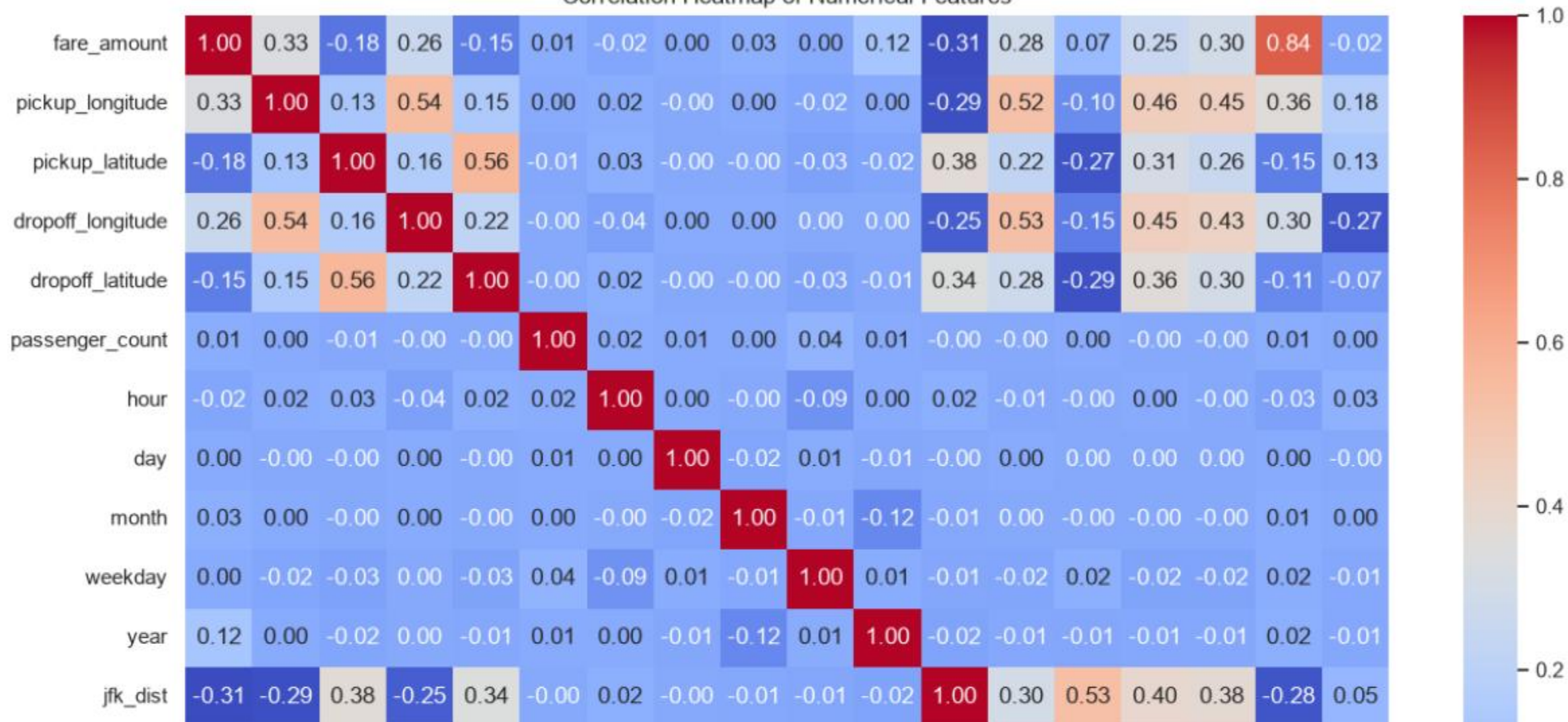
High-fare outliers elevate the mean.

QUESTION 10

Which features relate most to Fare?

Insight: Distance holds the strongest correlation (0.838). Other numerics demonstrate extremely weak linear association.

Correlation Heatmap of Numerical Features



QUESTION 11

When is ride demand highest?

Insight: Peak concentrations appear consistently during evening hours (7 PM–11 PM) across most days.



Key EDA Findings

01 / PRIMARY DRIVER

Trip Distance

Strongest observed linear indicator of fare amount.

02 / CATEGORICALS

Weak Influencers

Passenger count, car condition, traffic, and weather show minimal fare variance.

03 / FINANCIAL METRICS

Skewed Distribution

Fares heavily concentrated below \$20; high-fare outliers exist.

04 / TEMPORAL TRENDS

Diverging Peaks

Highest average fares in early morning; highest demand in evening.

Conclusion

1

Clear Foundations

Successfully extracted core insights into fare drivers and ride activity patterns.

2

Key Predictor

Distance identified as the primary feature
categoricals show minimal linear impact.

3

Ready for ML

Dataset is fully cleaned and prepped for the predictive modeling stage.

Thank You

Questions?